

this is probably one of the best and most expensive solutions available right now.

Another firm, Quanergy, recently announced a solid-state LiDAR named S3 that has no moving parts or micro-mirrors. Instead, it uses an optical-phased array as a transmitter, which can steer pulses of light by shifting the phase of a laser pulse as it is projected through the array. S3 is due to be available in September.

The major benefit of using LiDAR systems is that the data these generate is easier to compute and

the cyclist, but LiDAR laser picks it up, calculates the distance and prevents the car from moving ahead until the cyclist has passed.

Cameras. Mercedes F015, about which we wrote last year, does not use LiDAR, but a mix of 3D stereoscopic cameras, radars and ultrasonic sensors for its autonomous functions. It is very much possible to set up driver-assistance systems using just camera data, like how Land Rover Discovery Sport does with a Bosch stereo video camera. Its spatial measurements and video signals provide enough data to

PathFindIR is a compact thermal imaging camera that significantly reduces the hazards of night time driving, and is designed for both military vehicles as well as commercial vehicles as an aftermarket part. These systems have improved from mere night vision to being able to identify a hazardous object or even go as far as alerting the object.

Mercedes had CL550 a few years back that would identify pedestrians on the road and flash them with four blasts of diverted headlight while highlighting them with white brackets on users' screens.

While LiDAR and cameras are great for generating an accurate map of the car's surroundings, these are not ideal for monitoring the speed of other moving objects in real-time. So what do we do?

Sense of hearing

Just like Tesla, which uses radar as one among its suite of electronic senses with which the car finds its way around town, there are other vendors too who plan to use automotive radar.

Radar. CMOS seems to be the thing in the world of automotive radar. NXP has millimeter-wave technology for its automotive radar, which is based on a digital CMOS process that was not believed possible. The solution covers 76GHz to 81GHz range.

Imec and Vrije Universiteit Brussel from Belgium announced some time back what they then claimed to be the world's first 79GHz radar transmitter based on digital 28nm CMOS. "With an output power above 10dBm, the transmitter front-end paves the way for full radar-on-chip solutions for automotive and smart environment applications," stated the release.

Freescall has a 77GHz packaged radar front-end chipset but

Crash under a brightly-lit sky

A fatal crash occurred late June this year, when a Tesla Model S running in its almost autonomous autopilot mode rammed into a trailer lying perpendicular on the road.

Tesla's statement says, "Neither autopilot nor the driver noticed the white side of the tractor-trailer against a brightly-lit sky, so the brake was not applied. The high ride height of the trailer combined with its positioning across the road and the extremely-rare circumstances of the impact caused Model S to pass under the trailer, with the bottom of the trailer impacting the windshield of the car."

Is India ready

Indian roads are in a unique situation when compared to those in other countries. While some of our roads feature full road markings and signage, others make you think whether the road you are driving on was borrowed from Harappa or Mohenjo Daro archaeological sites.

A good number of development teams sit in India working on technology that can help in the building of self-driving cars, but a few, if any, car manufacturers believe that our roads are not ready to take autonomous vehicles—at least for a few decades!

process as compared to the data generated from a camera. These can also work in places where GPS and mapping data are not really accurate. Volvo Group has an autonomous FMX heavy-duty vehicle designed to work in underground mines without the need for a driver.

Of all the demonstrations that we at EFY have come across of LiDAR usage, perhaps the best is when the car picks up a cyclist waiting to cross the road on the other side of a pickup truck. Drivers in the car are not able to see

calculate, for example, the distance to vehicles ahead.

This is nothing new, of course, as Subaru has been marketing their own version since 2010, and Continental has their own forward-looking braking system with stereoscopic cameras that is just as old.

Infrared or thermal imaging cameras have been available that can detect hidden objects like people or animals in total darkness, smoke or fog-like conditions when normal detection systems may not easily pick these up. FLIR Systems